A · PATIENT DEMOGRAPHICS & INJURY PROFILE — INTAKE

Patient Name _____ Age _____ Sex ☐ M ☐ F Date _____

Sport / Occupation _____ Level ☐ Rec ☐ Comp ☐ Elite Dominant ☐ L ☐ R Side ☐ L ☐ R ☐ Bilat

Date of Injury _____ Referrer _____ Prior sprains # _____ Imaging _____

Mechanism ☐ Inversion (PF + Inv) ☐ Eversion ☐ Hyper-DF / ER ☐ Axial Plantar Load ☐ Non-contact ☐ Direct Blow ☐ Insidious / Overuse

NPRS Now (circle) 0 1 2 3 4 5 6 7 8 9 10 Worst 24 h _____ Best 24 h _____ Activity _____

B · EMERGENCY SCREENING & RED FLAGS — COMPLETE BEFORE ANY LOADING OR MANUAL THERAPY — TRIAGE GATE

OTTAWA EXCLUSIONS: ☐ Age < 5 ☐ Intox ☐ Altered mental ☐ Distracting inj ☐ Gross swelling ☐ Neuropathy ☐ > 10 days

MALLEOLAR ZONE → ANKLE RADIOGRAPH [H] MDCALC — SN 98%

- ☐ Post. edge / tip of **lateral malleolus** (distal 6 cm)
- ☐ Post. edge / tip of **medial malleolus** (distal 6 cm)
- ☐ **Unable to bear weight 4 steps** (immediately & at assessment)

☐ Negative → No Ankle X-ray ☐ Positive → Ankle Radiograph

MIDFOOT ZONE → FOOT RADIOGRAPH [H] — SN 98%

- ☐ Bone tenderness at **base of 5th metatarsal**
- ☐ Bone tenderness at **navicular bone**
- ☐ **Unable to bear weight 4 steps** (immediately & at assessment)

☐ Negative → No Foot X-ray ☐ Positive → Foot Radiograph

HIGH-RISK FRACTURE CHECKLIST — DO NOT MISS

- ☐ **Lisfranc** [L*] _ ☐ Plantar ecchymosis ☐ Piano-key ☐ 1st–2nd MT diastasis
- ☐ **Zone 2 Jones Fx** (watershed non-union) • **Maisonneuve** (palpate fibular neck)
- ☐ **Calcaneal Stress Fx** ((+) Squeeze) • **Talar Dome / OCL** (joint pain in full PF)
- ☐ **SKELETALLY IMMATURE?** → distal fibular tenderness = **Salter-Harris**, not sprain

NON-ORTHOPAEDIC RED FLAGS — SYSTEMIC

- ☐ **Wells DVT:** ☐ ≥ 2 (Duplex US) • ☐ < 2 (Unlikely) · Score: _____ MDCalc
- ☐ **Compartment Syndrome — 6 Ps:** ☐ Pain on stretch ☐ Paresthesia ☐ Pallor ☐ Paralysis ☐ Pulseless ☐ Poikilothermia

ΔP (Diastolic BP − Compartment Pressure) = ___ mmHg → ≤ 30 mmHg = EMERGENCY FASCIOTOMY INDICATED

CHARCOT NEUROARTHROPATHY — NEUROPATHIC + HOT RED FOOT — EMERGENCY

☐ Temp diff > 2 °C ☐ Erythema reduces on 10 min elevation ☐ Minimal/absent pain ☐ Palpable pulses present

→ IMMEDIATE total-contact offloading + urgent referral. NO loading. NO manual therapy. NO ESWT.

- ☐ Septic Arthritis: hot effusion · T > 38 °C · synovial WBC > 50k
- ☐ Acute Gout: 1st MTP erythema, exquisite pain, hx / hyperuricaemia
- ☐ **Budapest CRPS** [H] Calc (≥ 3/4 Sx, ≥ 2/4 Signs): ☐ Sensory ☐ Vasomotor ☐ Edema ☐ Trophic → ☐ I ☐ II
- ☐ **Necrotizing fasciitis:** pain out of proportion, rapid, systemic toxicity, crepitus → **ED NOW**

ACHILLES RUPTURE EMERGENCY TRIAD — ≥ 2 POSITIVE (SN/SP > 98%)

☐ **Thompson** • ☐ **Palpable Gap (2–6 cm)** ☐ **Matles Test** (prone 90°)

Gap may be obscured by haematoma after 24–48 h; negative late palpation does not exclude rupture.

≥ 2 POSITIVE: Equinus boot + urgent orthopaedic referral (< 48 h). No loading.

TRIAGE DISPOSITION: ☐ **Cleared — proceed** ☐ Imaging referral ☐ Urgent orthopaedic ☐ **Emergency Department NOW**

C · PATHWAY ROUTER & FAST-TRACK ASSESSMENT (< 2-MINUTE SCREEN) — RAPID

CLINICAL PATHWAY ROUTER — Tick indicated route → navigate to page

☐ A · SIMPLE: 1 → 6 → 7 → 9 ☐ B · COMPLEX: 1 → 2 → 3 → 4 → 6 → 7 → 8 → 9 ☐ C · POST-OP: 1 → 5 → 6 → 7 → 8 → 9

Heel Walk (L4/L5)	Toe Walk (S1)	Single-Leg Stance 10 s	Figure-of-8 Edema (cm)
L ☐ P ☐ F R ☐ P ☐ F	L ☐ P ☐ F R ☐ P ☐ F	L ☐ P ☐ F R ☐ P ☐ F	L _____ R _____ Δ _____

Fast Classification: ☐ Simple Lateral Sprain Gr I / II ☐ Mild Plantar Fasciopathy ☐ Minor Overuse

D · 24-HOUR SYMPTOM BEHAVIOUR, PAIN MAP & PATIENT GOALS — SUBJECTIVE

MORNING PAIN & STIFFNESS

- ☐ **First-step heel pain**, eases < 10–15 min → Plantar fasciopathy
- ☐ **Achilles stiffness**, warms up with activity → Tendinopathy
- ☐ **Stiffness > 1 h** → Inflammatory (SpA / RA / gout)

Duration _____ min · Gel ☐ Y ☐ N

FOOTWEAR & ORTHOTICS

Type _____ Orthotics ☐ No ☐ OTC ☐ Cust _____ km

_____ Drop _____ mm

AGGRAVATING SURFACES

☐ Hard ☐ Uneven ☐ Stairs ☐ Inclines ☐ Sand ☐ Barefoot ☐ Cutting

Night pain ☐ Mechanical ☐ **Unremitting**

PMH / Risk: ☐ Diabetes / Neuropathy ☐ Inflammatory arthritis ☐ Fluoroquinolones ☐ Corticosteroid injection ☐ Smoker ☐ Hypermobility ☐ Anticoagulants

PAIN MAP — shade symptoms · X = max tenderness · circle side: L R



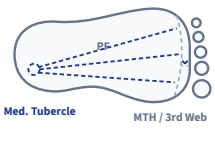
LATERAL



MEDIAL



ANTERIOR



PLANTAR

PATIENT GOALS · PSFS (0–10)

1. _____ _____ /10
2. _____ _____ /10
3. _____ _____ /10

Clinician _____ Signature _____

Unbiased Differential Diagnosis & Neurological Matrix

Patient _____

Bias-Free Decision Architecture: Screen **every** compartment before committing to a hypothesis. Tick all **suspected** conditions, then rank the top 3 in the Working Hypothesis box. Mark L/R where relevant.

1 · LATERAL ANKLE — 5 TARGETS

- ☐ **Acute Lateral Sprain** — ☐ Gr I micro ☐ Gr II partial ☐ Gr III complete rupture
- ☐ **Peroneal Tendon** — ☐ Subluxation ☐ Longitudinal split ☐ Pain resisted Ev in DF — BJSM Guideline 2018
- ☐ **Sinus Tarsi Syndrome** — palpation 1 cm anterior to lateral malleolus; relief with lidocaine
- ☐ **Anterolateral Impingement** — (+) Molloy-Bendall sign (pressure + forced DF)
- ☐ **Lateral Process of Talus Fx** — "snowboarder's"; tender just distal to fibular tip

2 · MEDIAL ANKLE — 5 TARGETS

- ☐ **Deltoid Ligament Sprain** — Eversion stress / Kleiger ER; **palpate fibular neck**
- ☐ **PTTD Staging** — ☐ I Tenosynovitis ☐ II Flexible flatfoot / "Too-Many-Toes" ☐ III Rigid pes planus ☐ IV Ankle valgus
- ☐ **Tarsal Tunnel Syndrome** — (+) Dorsiflexion-Eversion test + Tinel at flexor retinaculum
- ☐ **Anterior Tarsal Tunnel** — deep peroneal n., 1st web space numbness, (+) Tinel over inf. extensor retinaculum
- ☐ **FHL Tendinopathy** — (+) Pseudo-Hallux Rigidus test (1st MTP ext in ankle DF vs PF)

3 · ANTERIOR ANKLE — 2 TARGETS

- ☐ **Syndesmotic High Ankle Sprain** — ☐ AITFL ☐ PITFL ☐ Kleiger ER ☐ Hopkins Squeeze
  Interosseous tenderness height _____ cm above joint line (↑ height = ↑ time to RTS)
- ☐ **Anterior Impingement** — ☐ Bony (tibiotalar osteophytes) ☐ Synovial meniscoid

4 · POSTERIOR ANKLE — 4 TARGETS

- ☐ **Midportion Achilles** — 2–6 cm watershed; ☐ Royal London (+ in DF) ☐ Arc sign
- ☐ **Insertional Achilles** — ☐ Enthesopathy ☐ Haglund's ☐ 2-Finger Pinch (bursa)
- ☐ **Os Trigonum / Posterior Impingement** — pain on forced PF
- ☐ **Sever's Apophysitis** — adolescents; (+) calcaneal squeeze

5 · PLANTAR & MIDFOOT — 7 TARGETS

- ☐ **Plantar Fasciopathy** — medial calcaneal tubercle; (+) Windlass test
- ☐ **Baxter's Nerve Entrapment** — medial abductor hallucis border; ADM weakness; **no** start-up pain
- ☐ **Medial Calcaneal Nerve** — burning medial heel, Tinel (+)
- ☐ **Morton's Neuroma** — Mulder's click; 3rd web space
- ☐ **Navicular Stress Fracture** — **"N-spot" tenderness** (dorsal proximal navicular)
- ☐ **Accessory Navicular** — prominent medial navicular, tender synchondrosis
- ☐ **Cuboid Syndrome** — lateral midfoot, plantar cuboid tender, post-inversion

WORKING HYPOTHESIS RANKING — SYNTHESIS

1° _____ Conf. _____ %

2° _____ Conf. _____ %

3° _____ Conf. _____ %

Must-not-miss excluded? ☐ Yes ☐ No

Irritability ☐ Low ☐ Mod ☐ High

Mechanism ☐ Nociceptive ☐ Neuropathic ☐ Nociplastic

☐ **Tarsal Coalition** — adolescent, recurrent sprains, rigid flatfoot, peroneal spasm, loss of subtalar motion

SPINAL & KINETIC CHAIN DIFFERENTIAL SCREENING — LUMBAR L4–S1 NEURO-MAPPING — NEURO

LEVEL	MYOTOME (KEY MUSCLE)	REFLEX	DERMATOME	MYOTOME L / R (0–5)	REFLEX L / R (0–4+)	SENSATION L / R
L4	Tibialis anterior (DF / inversion)	Patellar	Medial leg & medial malleolus	L ___ R ___	L ___ R ___	L ___ R ___
L5	Extensor hallucis longus	Medial hamstrings	Dorsum foot / 1st web space	L ___ R ___	L ___ R ___	L ___ R ___
S1	Peroneals & Gastrocnemius	Achilles	Lateral foot & sole	L ___ R ___	L ___ R ___	L ___ R ___

Grading key: Myotome ≤ 4/5 deficit | Reflex: 0-1+ hypo / ≥ 3+ hyper | Sensation: Intact / Impaired / Absent

NEURAL TENSION & DISTAL BIASING — SLR (+) RANGE 35–70°

TEST	SEQUENCE / BIAS	ONSET ANGLE L/R (°)	FAMILIAR SX	STRUCTURAL DIFF. (+)
Straight Leg Raise	Hip flex, knee ext; positive 35–70°	L ___ R ___	☐	☐
TED — Tibial	SLR + ankle DF + Eversion (+ toe ext)	L ___ R ___	☐	☐
SID — Sural	SLR + ankle DF + Inversion	L ___ R ___	☐	☐
PIP — Peroneal	SLR + ankle PF + Inversion	L ___ R ___	☐	☐
Slump Test	Thoracic + cervical flex, knee ext, DF → release neck	L ___ R ___	☐	☐ Sx ↓ on neck ext

Kinetic chain: ☐ Lumbar ROM reproduces Sx ☐ Hip abductor weakness ☐ Dynamic knee valgus ☐ Hip IR deficit

⛔ ☐ **CAUDA EQUINA RED FLAG — saddle anaesthesia · bladder / bowel dysfunction · bilateral leg signs · sexual dysfunction → EMERGENCY MRI / ED same day**

DIFFERENTIAL REASONING NOTES · IMAGING FINDINGS

Clinician _____ **Date** _____

QUANTITATIVE MOBILITY GRID — LEFT VS RIGHT

ROM · ALIGNMENT

MEASURE	NORMATIVE / THRESHOLD	LEFT	RIGHT	Δ / LSI	FLAG / INTERPRETATION
WBLT Knee-to-Wall — Distance (cm)	≥ 10–12.5 cm · **Asym > 1.5 cm**				☐
WBLT — Inclinometer (°)	35–45° · **Asym > 4.7°**				☐
Silfverskiöld — DF Knee 0° (°)	Gastrocnemius + soleus length				☐ Gain ≥ 10° = **Isolated Gastrocnemius**
Silfverskiöld — DF Knee 90° (°)	Soleus / capsular length				☐ Gain < 5° = **Soleus / Capsule**
Silfverskiöld Gain (90° – 0°)	Gain ≥ 10° (Gastroc) · Gain < 5° (Soleus/Capsule)				
Subtalar Inversion (°)	20–30°				Inv : Ev ratio **2 : 1** L ____ R ____
Subtalar Eversion (°)	10–15°				
Calcaneal Glides (Med / Lat)	Hypo · Normal · Hyper	↓ N ↑	↓ N ↑		☐
Midtarsal (Chopart) Mobility	Hypo · Normal · Hyper	↓ N ↑	↓ N ↑		☐
1st MTP Extension — NWB (°)	≥ 60–65° gait · 70–90° sprinting				☐ **Functional Hallux Limitus** (≥ 60° NWB, ↓ WB)
1st MTP Extension — WB (°)	Jack's / Hubscher manoeuvre				☐ **Hallux Rigidus** (< 30°)
Foot Posture Index (FPI-6) · FPI Guide	−12 to +12 · Neutral 0 to +5 · Pron +6 to +9 · Highly pronated ≥ +10				☐ Sup ☐ Neu ☐ Pron ☐ High
Navicular Drop Test (mm)	Normal 6–10 mm · **Hyperpronation > 10 mm**				☐

Measurement error: goniometry ± 5° – inclinometer ± 2–3° — differences smaller than this are not clinically meaningful.

MUSCLE CAPACITY & STRENGTH QUANTIFICATION

HHD · MMT · ENDURANCE

SINGLE-LEG HEEL RAISE TEST — 60 BPM

MALES 32–37 · FEMALES 27–31 · ATHLETES ≥ 38–45+ REPS

SIDE	REPETITIONS	MAX HEIGHT (CM)	TOTAL WORK (REPS × CM)	LSI %	TERMINATION REASON
Left					☐ Fatigue ☐ Pain ☐ Cadence ☐ Height ↓
Right					☐ Fatigue ☐ Pain ☐ Cadence ☐ Height ↓

HAND-HELD DYNAMOMETRY (N · N/KG) & MMT

BODY MASS ____ KG

MUSCLE	TEST POSITION	L (N)	L (N/KG)	R (N)	R (N/KG)	MMT L / R	LSI %
Tibialis Posterior	Inversion in plantarflexion					____ / ____	
Peroneus Longus	1st ray PF + eversion					____ / ____	
Peroneus Brevis	Pure eversion vs lateral 5th MT					____ / ____	
Tibialis Anterior	DF + inversion					____ / ____	
FHL	Hallux IP flexion, ankle neutral					____ / ____	

SYSTEMATIC PALPATION GRID — 14 ANATOMICAL TARGETS

TICK TENDER SIDE

LATERAL	L · R	MEDIAL	L · R	ANTERIOR	L · R	PLANTAR	L · R
1. ATFL	☐ ☐	6. Deltoid ligament	☐ ☐	10. AITFL	☐ ☐	13. Med. calc. tubercle	☐ ☐
2. CFL	☐ ☐	7. Tib. posterior tendon	☐ ☐	11. Ant. joint line	☐ ☐	14. 3rd web space	☐ ☐
3. PTFL	☐ ☐	8. Tarsal tunnel	☐ ☐	12. Sinus tarsi	☐ ☐		
4. Peroneal tendons	☐ ☐	9. Navicular / N-spot	☐ ☐				
5. Base of 5th MT	☐ ☐						

Effusion 0 1+ 2+ 3+

GAIT · OBSERVATION · BIOMECHANICAL NOTES

Clinician _____ Date _____

Interactive references: Underline Atlas card · Underline + · Verified video · Calc digital calculator · Guide CPG & consensus

Evidence Certainty: [H] Multi-study validated · [M] Moderate · [L] Single / unreplicated (*flagged) · [E] Expert consensus

LATERAL LIGAMENT — VAN DIJK DAY-5 CLUSTER [M*]

DELAYED EXAM

Sn 100% · Sp 77% · +LR 4.35 · −LR ≈ 0.00

- ☐ **ATFL palpation pain**
- ☐ **Lateral haematoma** / discolouration
- ☐ **Anterior Drawer** (+) at day 4–5
- ☐ Adjunct: **Talar Tilt** (CFL)

Cluster ☐ (+) ATFL rupture likely ☐ (−)

SYNDESMOSIS — SMAN CLUSTER [L*]

HIGH ANKLE

Sp 96% · OR 13.8 · Refer MRI if (+)

- ☐ **AITFL tenderness**
- ☐ **DF-External Rotation (Kleiger)** (+)
- ☐ **Squeeze test (Hopkins)** (+)
- ☐ Adjunct: **Fibular Translation**

Cluster ☐ (+) Syndesmotic injury ☐ (−)

ACHILLES MIDPORTION CLUSTER [M*]

TENDINOPATHY

Sn 84% · Sp 83% · +LR 4.94 [M*] · −LR 0.19

- ☐ **Palpable thickening** (2–6 cm proximal to insertion)
- ☐ **Localised tenderness** on palpation
- ☐ **(+) Royal London Hospital** *or* **(+) Arc sign**

Cluster ☐ (+) Midportion AT ☐ (−)

PLANTAR HEEL PAIN CLUSTER [H]

HEEL

WB Windlass Sp 100% [H] · Rule-in test

- ☐ **Medial calcaneal tubercle tenderness**
- ☐ **(+) Weight-bearing Windlass** [H]
- ☐ **Absence** of DF-Eversion neurodynamic pain (excludes TTS / Baxter's)

Cluster ☐ (+) Plantar fasciopathy ☐ (−)

LISFRANC CLUSTER — TARSOMETATARSAL COMPLEX [L*]

SURGICAL RISK

Sn / Sp > 90% [L*] · WB X-ray / CT if (+)

☐ **2nd MT base tenderness** ☐ **Midfoot abduction stress** (+) ☐ **Plantar ecchymosis**

☐ (+) ☐ (−) **C1–M2 gap** _____ mm

ADDITIONAL INDIVIDUAL TESTS — RESULT LOG

L / R · + / −

TEST	L	R	TEST	L	R	TEST	L	R
Anterolateral Drawer			Thompson			Mulder's Click		
Eversion Stress (Deltoid)			Single-Leg Hop (pain)			Tinel — Tarsal Tunnel		
Peroneal Subluxation			Too-Many-Toes			Calcaneal Squeeze		
Cotton (Lateral Talar Shift)			SL Heel Rise (PTTD)			Forced PF (Posterior)		

STANDARDISED PATIENT-REPORTED OUTCOME MEASURES (PROMS) DASHBOARD [H]

MCID · RTS CUTOFFS

MEASURE	RANGE	MCID [H]	CLEARANCE / RTS CUTOFF	BASE	WK 4	WK 8	WK 12	D/C	≥ MCID
FAAM-ADL FAAM Calc	0–100%	8 pts [H]	—						☐
FAAM-Sports	0–100%	9 pts [H]	RTS ≥ 90–95%						☐
VISA-A VISA-A Calc	0–100 pts	12–14 pts [H]	RTS ≥ 85–90 pts						☐
CAIT CAIT Calc	0–30 pts	—	≤ 24 Instability [H] / ≥ 27–28 Stable						☐
LEFS LEFS Calc	0–80 pts	9 pts [H]	Clearance ≥ 72–75						☐

< MCID at 6 weeks → re-screen Section B red flags, reconsider diagnosis, consider imaging or referral.

CLINICAL DIAGNOSIS · ICD-10 · PROGNOSIS & SMART GOALS

Clinician _____ Date _____

Universal Progression Gate (all phases): wound healed / no infection · Pain ≤ 2/10 at rest | Effusion ≤ 1+ · no DVT signs · surgeon-specific restrictions **override** this template. Time frames are **minimums** — progress on **criteria**, not calendar.

VTE prophylaxis: ☐ prescribed ☐ not indicated ☐ surgeon-directed | Re-screen Wells at each visit while immobilised

1 · LATERAL LIGAMENT — BROSTRÖM VS INTERNALBRACE — OJSM LIGAMENT 2020 ☐ A ☐ B

PARAMETER	A · ISOLATED MODIFIED BROSTRÖM-GOULD	B · BROSTRÖM + INTERNALBRACE AUGMENTATION	MET
Splint / Boot	Splint 0–2 wks → CAM boot 2–6 wks	Splint 0–10/14 days → CAM boot to wk 4–6	☐
Weight-Bearing	NWB 0–2 wks → PWB 2–4 wks → FWB in boot	**WBAT in boot from day 10–14**	☐
ROM Restrictions	**Inversion PROHIBITED 0–6 wks · Combined Inversion + Plantarflexion PROHIBITED 0–12 wks.** Sagittal DF / PF per surgeon.		☐
Shoe Weaning	Weeks 6–8 (unaugmented) + lace-up brace	Weeks 4–6 (augmented) + lace-up brace	☐
Progression Criteria	Normal gait without boot · SL balance 30 s · WBLT within 1.5 cm → jog; Hop LSI ≥ 90% + CAIT ≥ 27 → RTS.		☐

2 · ACHILLES TENDON REPAIR — OPEN VS MIS — OJSM ACHILLES 2019 ☐ OPEN ☐ MIS

PARAMETER	OPEN REPAIR	PERCUTANEOUS / MIS REPAIR	MET
Boot Set-up	CAM boot with **3 heel wedges (3 × 1 cm ≈ 30° equinus)**; protected WB progressing to FWB in boot per surgeon.		☐
Wedge Reduction	Remove 1 wedge every **2 wks**	Remove 1 wedge every **10–14 days**	☐
Neutral (Plantigrade)	Reached in boot by **week 6–8**. Wedge removal dates: W1 _____ W2 _____ W3 _____		☐
Shoe Weaning	Transition with **5–10 mm heel lift in BOTH shoes** (avoids pelvic obliquity); wean lift over 2–4 wks.		☐

CRITICAL RED LINE: NO passive/active dorsiflexion past 0° (plantigrade) until weeks 8–10. No calf stretching. Tendon elongation is irreversible and permanently weakens push-off.

3 · SYNDESMOTIC FIXATION — RIGID SCREWS VS DYNAMIC TIGHTROPE ☐ SCREW ☐ TIGHTROPE

PARAMETER	RIGID TRANS-SYNDESMOTIC SCREWS	DYNAMIC SUTURE-BUTTON (TIGHTROPE)	MET
Weight-Bearing	**Strict NWB 0–6 wks**	TDWB 0–2 wks → PWB 2–3 wks → **FWBAT 3–4 wks**	☐
Hardware	Removal at 8–12 wks (before full impact)	No routine removal	☐
Restrictions	**Prohibit External Rotation + Eversion for 6 wks** (both techniques). Avoid aggressive end-range DF loading early.		☐

4 · TALAR OSTEOCHONDRAL LESIONS — BMS / MICROFRACTURE VS OATS ☐ BMS ☐ OATS

PARAMETER	BONE MARROW STIMULATION / MICROFRACTURE	OSTEOCHONDRAL AUTOGRAFT TRANSFER (OATS)	MET
Lesion Profile	**Defect:** ☐ Contained ☐ Uncontained | **Size:** _____ cm² | **Plug:** Ø _____ mm | **Osteotomy:** ☐ Yes ☐ No		☐
Weight-Bearing	☐ Small/Contained (< 1 cm²): **PWB @ 2 wks** ☐ Large/Uncontained (≥ 1–1.5 cm²): **Strict NWB 4–6 wks** (protects clot)	**Strict NWB 0–6 wks** (protects donor plug & medial malleolar osteotomy)	☐
Early Motion	Early CPM 0–30° (fibrocartilage stimulation)	Gentle AROM once osteotomy stable	☐
Impact / Running	Impact at **3–4 months**	Impact at **5–6 months** post CT-confirmed union	☐

SURGEON-SPECIFIC INSTRUCTIONS · OPERATIVE FINDINGS · PHASE LOG

Clinician _____ Post-op Week _____

EXTRACORPOREAL SHOCKWAVE THERAPY (ESWT) · ISMST INDICATIONS

3–5 SESSIONS · 7-DAY INTERVALS

RX	INDICATION	TYPE	ENERGY	PULSES	FREQ	PROBE / TECHNIQUE
☐	**Plantar Fasciopathy**	fESWT	**0.15–0.28 mJ/mm²**	2000	4–8 Hz	Medial tubercle; coupling gel; patient-guided focus
☐	**Plantar Fasciopathy**	rESWT	**2.0–3.5 Bar**	2000	8–10 Hz	Circular sweep over origin & proximal fascia
☐	**Achilles Midportion**	rESWT	**2.0–3.5 Bar**	2000	8–10 Hz	Probe 90° to tendon; **pinch-grip** stabilisation
☐	**Achilles Insertional**	rESWT	**1.5–2.2 Bar**	2000	8–10 Hz	Probe 30–45° cephalad; **no perpendicular cortical impact**

DRUG & LOAD INTERLOCK: Avoid routine NSAIDs during the loading phase where possible (mechanistic concern re: tenocyte mechanotransduction — evidence limited; prescribing decisions rest with the referring physician). NO ballistic loading 48 h. Begin isometrics 5 × 45 s at 48–72 h.
FLUOROQUINOLONE EXPOSURE (current or within 6 months) → ESWT and heavy tendon loading CONTRAINDICATED. Tendon rupture risk persists post-course.

CLASS IV LASER (HILT)

EYEWEAR OD5+

λ	PRIMARY MECHANISM
810 nm	Cytochrome c oxidase → ↑ ATP
980 nm	H_2O / Hb absorption → thermal wash-out
1064 nm	Deep tissue optical window

RX	PHASE	MODE	POWER	DOSE
☐	**Acute sprain**	Pulsed	2–4 W	4–8 J/cm²
☐	**Chronic tendon**	CW	6–12 W	15–30 J/cm²

CI: Malignancy, pregnancy (abdomen), photosensitising drugs, tattoos, eyes.

TECAR THERAPY (448 KHZ)

CET · RET

MODE	ELECTRODE	TARGET TISSUE
CET	Insulated (capacitive)	Vascular, muscle, edema
RET	Conductive (resistive)	Dense collagen, tendon, capsule

RX	LEVEL	INTENSITY	PHASE
☐	**Athermic**	5–15%	Acute 0–72 h
☐	**Mild thermal**	20–35%	Subacute
☐	**Hyperthermic**	55–75%	Chronic scar remodelling

CI: Pacemaker, pregnancy, DVT, malignancy, sensory loss.

CRYOTHERAPY PROTOCOL

ACUTE · POST-EXERCISE

Indication: Acute pain & edema, post-loading flare • **Dose / Duration:** Crushed ice pack / cuff 15–20 min, barrier cloth, repeat q2–3h

CI: Cold urticaria, Raynaud's, cryoglobulinaemia, sensory loss, peripheral vascular disease (PVD).

ELECTROTHERAPY — TENS · IFC · NMES · HVPC · IONTOPHORESIS

RX	GOAL	CURRENT	FREQUENCY	PULSE / CARRIER	INTENSITY	DOSE	EVIDENCE
☐	**Pain — acute / during activity**	Conventional TENS	80–120 Hz	50–100 µs	Strong but comfortable, sub-motor	30–60 min; can run during exercise	[M]
☐	**Pain — chronic / deep**	Acupuncture-like TENS	2–4 Hz	150–250 µs	Visible rhythmic contraction	20–30 min	[M]
☐	**Prevent tolerance**	Mixed-frequency TENS	Alternating 4 / 100 Hz	—	Re-titrate intensity mid-session	Daily use	[M]
☐	**Pain — deep tissue**	IFC quadripolar	Beat 80–150 Hz acute · 1–10 Hz chronic	Carrier 4000 Hz	Strong comfortable	15–20 min, sweep on	[L]
☐	**Strength / post-op inhibition**	NMES biphasic symmetric	30–50 Hz	200–400 µs	Max tolerable (target ≥ 50% MVIC)	10–15 s on / 30–50 s off · 10–20 contractions · 3×/wk · superimposed on voluntary contraction	[H]
☐	**Strength (alternative)**	Russian current	50 Hz bursts	Carrier 2500 Hz, 10 ms on/off	Max tolerable	10 s on / 50 s off · 10 reps	[M]
☐	**Edema — acute**	HVPC negative polarity	100–120 Hz	Twin-peak monophasic	Sub-motor	30 min, first 24–72 h, limb elevated	[L]
☐	**Edema — muscle pump**	Motor-level NMES	1–10 Hz	200–300 µs	Rhythmic contraction	20–30 min, limb elevated	[M]
☐	**Plantar fasciopathy**	Iontophoresis (dexamethasone)	DC	—	2–4 mA	40–80 mA·min · 6 sessions / 2 wks	[M]

Intensity is the dominant TENS parameter — under-dosing is the commonest failure. · Daily identical TENS produces tolerance in ~5 days: vary frequency or use mixed mode. · NMES works best superimposed on voluntary contraction, not as a replacement.

CI: pacemaker / ICD · carotid sinus / anterior neck · pregnancy (abdomen, low back) · over DVT · active malignancy in field · broken skin · impaired sensation · epilepsy (head/neck) · transthoracic placement.

TREATMENT RESPONSE · ADVERSE REACTIONS · DOSAGE ADJUSTMENTS

Clinician _____ Date _____

MANUAL THERAPY & NEURODYNAMICS

MAITLAND · MULLIGAN

RX	TECHNIQUE	GRADE / DOSE
☐	**Talocrural A-P glide**	I–II pain · III–IV DF ↓
☐	**Subtalar lateral / medial glide**	III–IV (Ev ↓ / Inv ↓)
☐	**Distal tibiofibular glide**	Grades I–IV (fibular position)
☐	**WB lunge MWM + belt**	3 × 10, pain-free

NERVE	SLIDER (ACUTE / IRRITABLE)	TENSIONER (CHRONIC)
Tibial (TED)	☐	☐
Sural (SID)	☐	☐
Peroneal (PIP)	☐	☐
Dose	2–3 × 10–15 oscillations	3 × 5–10, 1–2 s end-range

DRY NEEDLING SAFETY MAPPING

DEPTH VARIES WITH BMI · US GUIDANCE ADVISED

RX	MUSCLE	NEEDLE	DEPTH	APPROACH & NEUROVASCULAR SAFETY
☐	**Gastrocnemius**	0.25–0.30 × 40–50 mm	1.5–3 cm	Prone; medial/lateral heads; avoid midline (sural n., small saphenous v.)
☐	**Soleus**	0.30 × 50–60 mm	2.5–4.5 cm	Medial/lateral to gastroc border; stay superficial to deep posterior compartment
☐	**Tibialis Posterior**	0.30 × 60–75 mm	4–6 cm	**Medial approach, angled 15–20° under the tibia** — protects posterior tibial NV bundle
☐	**Peroneals**	0.25 × 30–40 mm	1–2.5 cm	Distal to fibular neck (common peroneal n.); superficial peroneal n. in distal 1/3

Safety Screen: ☐ Anticoagulants / bleeding screened ☐ No active infection / lymphedema ☐ Vasovagal history screened

MANUAL RELEASE & CUPPING

RX	TECHNIQUE	PARAMETERS	EVIDENCE
☐	**IASTM**	30–45° stroke angle · light–moderate pressure · 30–120 s per region · 1–2×/wk	[L]
☐	**Active Release / Pin-and-Stretch**	Pin in shortened position → active lengthening · 3–5 passes	[L]
☐	**Plantar fascia / gastroc-soleus MFR**	Sustained pressure 90–120 s to release	[L]
☐	**Dry cupping — static**	Light–moderate suction · 5–10 min	[L]
☐	**Dynamic (sliding) cupping**	Lubricant · light suction · 3–5 min along fiber direction	[L]
☐	**Cupping with movement**	Cup in place + active ROM · 2–3 min	[L]
☐	**Wet cupping (Hijama)**	Skin-breaching: sterile single-use equipment, bloodborne-pathogen precautions. Confirm scope of practice & licensure before use.	[L]

CONSENT OBTAINED FOR: ☐ ESWT ☐ Dry needling ☐ Grade III–IV manual therapy ☐ Cupping (dry / wet) | Risks & alternatives discussed ☐ | Patient may decline ☐

CI: anticoagulants / bleeding disorders · fragile skin (elderly, chronic steroids) · over DVT / varicosities · open wounds · neuropathic / diabetic foot · active infection · pregnancy (abdomen). Wet cupping additionally: anaemia, immunosuppression.

TECHNIQUE RESPONSE · ADVERSE REACTIONS

Clinician _____ Date _____

TENDON LOADING CONTINUUM — COOK & PURDAM 4-STAGE MODEL

CONTINUUM

TICK CURRENT STAGE

NOW	STAGE	PARAMETERS	PROGRESSION CRITERIA
☐	1 · Isometrics	5 × 45 s holds @ 70% MVC (Rio analgesic protocol) · Knee 90° (Soleus) vs 0° (Gastroc)	Analgesia post-set; NPRS ≤ 3/10
☐	2 · Heavy Slow Resistance	Beyer 12-wk protocol · 3 s concentric / 3 s eccentric · Midportion off step · **Insertional strictly flat ground**	SLHR ≥ 25 reps; 24-h response stable
☐	3 · Energy Storage	Plyometrics: pogo hops, skipping, bounding, jump-landing → running progression	SLHR LSI ≥ 90%; hop pain ≤ 3/10
☐	4 · Sport-Specific	Cutting, deceleration, COD, sprint exposure & full-volume training	RTS battery LSI ≥ 90–95%

BEYER HSR 12-WEEK PROGRESSION & LOAD LOG

HSR PROTOCOL (ABSTRACT)

3 × 15 → 4 × 6 · 3 SESSIONS / WK

BLOCK	WK 1	WKS 2–3	WKS 4–5	WKS 6–8	WKS 9–12
Target RM × Sets	**15RM × 3**	**12RM × 3**	**10RM × 4**	**8RM × 4**	**6RM × 4**
Load (kg)					
Pain during / 24 h					

- GREEN 0–2/10 — Safe. Progress load.
- AMBER 3–5/10 — Acceptable if morning NPRS returns to baseline within 24 h.
- RED > 5/10 or 24-h flare — Regress one stage.

BLOOD FLOW RESTRICTION (BFR) TRAINING

BFR POSITION STAND 2019

AOP · DOPPLER

- ☐ **80% AOP Doppler calculation** (posterior tibial artery, supine, cuff on proximal thigh):
  AOP _____ mmHg × 0.80 = _____ mmHg working pressure
- ☐ **Active BFR:** 30-15-15-15 reps @ 20–30% 1RM · 30 s rest · cuff stays inflated (≤ 15–20 min max continuous)
- ☐ **Passive IPC (early post-op):** 4–5 cycles × 5 min ON / 5 min OFF @ 100% AOP

Cuff on PROXIMAL THIGH only — NEVER below the knee (common peroneal nerve).

Screen (exclude if ticked): ☐ DVT / VTE hx ☐ PAD ☐ Sickle cell ☐ Uncontrolled HTN ☐ Pregnancy ☐ Active infection / malignancy

RETURN-TO-RUNNING GATE (BEFORE STAGE 3 RUNNING)

ALL REQUIRED

☐ Pain-free brisk walk 30 min ☐ WBLT asymmetry ≤ 1.5 cm ☐ SL balance 30 s eyes open ☐ SLHR LSI ≥ 80% ☐ 10 pain-free SL hops (≤ 2/10) ☐ No next-morning flare

COMPREHENSIVE ATHLETIC RTS TESTING BATTERY

PASS: LSI ≥ 90–95%

TEST	PASS CRITERION	LEFT	RIGHT	LSI %	PASS
Hop Testing Battery					
Single Hop for Distance (cm)	LSI ≥ 90–95%				☐
Triple Hop for Distance (cm)	LSI ≥ 90–95%				☐
Crossover Hop (cm)	LSI ≥ 90–95%				☐
6-Meter Timed Hop (s)	LSI ≥ 90–95% (uninvolved ÷ involved)				☐
30 s Side Hop (reps)	Females ≥ 40–50 | Males ≥ 50–65 reps				☐
Dynamic Balance · Calf Endurance · Reactive Power					
Y-Balance — Anterior Reach (cm)	Asymmetry < 4 cm				☐
Y-Balance — Composite (% LL)	≥ 95% of limb length · LL _____ cm				☐
Single-Leg Heel Raise @ 60 bpm	♂ ≥ 32–37 · ♀ ≥ 27–31 · Ath ≥ 38–45 · LSI ≥ 90%				☐
Drop Jump 30 cm — GCT (ms)	< 250 ms				☐
Reactive Strength Index	RSI ≥ 2.0 (height m ÷ GCT s)				☐
Agility & Change of Direction					
T-Test (s)	Males < 9.5 s · Females < 10.5 s				☐
Pro-Agility 5-10-5 (s)	Males < 4.4 s · Females < 4.7 s				☐
5-0-5 Agility (s, turn L / R)	Asymmetry < 0.1 s				☐
Psychological Readiness					
TSK-11 TSK-11 Calc **or Ankle-RSI** Ankle-RSI	TSK-11 (≤ 17) or Ankle-RSI (≥ 75%)	TSK _____ RSI _____ %			☐

RTS Decision: ☐ Cleared — full RTS ☐ Modified training ☐ Not cleared

Tests passed _____ / 14

LSI caution: *the uninvolved limb also deconditions. Interpret against pre-injury or normative values where available.*

MOVEMENT QUALITY · LANDING MECHANICS · READINESS NOTES

Clinician _____ Test Date _____

Longitudinal Session-by-Session Tracking Flowsheet

Patient _____ Dx _____

#	DATE	NPRS R/A	AM PAIN	WBLT (CM)	SLB / SLHR	MODALITIES DELIVERED	MANUAL & NEURO APPLIED	LOAD & STAGE	ADHERENCE	INIT
1						ESWT | HILT | TECAR | BFR | Cryo | ES | Cup	Maitland | Mulligan | DN | IASTM	St 1 | St 2 | St 3 | St 4		
2						ESWT | HILT | TECAR | BFR | Cryo | ES | Cup	Maitland | Mulligan | DN | IASTM	St 1 | St 2 | St 3 | St 4		
3						ESWT | HILT | TECAR | BFR | Cryo | ES | Cup	Maitland | Mulligan | DN | IASTM	St 1 | St 2 | St 3 | St 4		
4						ESWT | HILT | TECAR | BFR | Cryo | ES | Cup	Maitland | Mulligan | DN | IASTM	St 1 | St 2 | St 3 | St 4		
5						ESWT | HILT | TECAR | BFR | Cryo | ES | Cup	Maitland | Mulligan | DN | IASTM	St 1 | St 2 | St 3 | St 4		
6						ESWT | HILT | TECAR | BFR | Cryo | ES | Cup	Maitland | Mulligan | DN | IASTM	St 1 | St 2 | St 3 | St 4		
7						ESWT | HILT | TECAR | BFR | Cryo | ES | Cup	Maitland | Mulligan | DN | IASTM	St 1 | St 2 | St 3 | St 4		
8						ESWT | HILT | TECAR | BFR | Cryo | ES | Cup	Maitland | Mulligan | DN | IASTM	St 1 | St 2 | St 3 | St 4		
9						ESWT | HILT | TECAR | BFR | Cryo | ES | Cup	Maitland | Mulligan | DN | IASTM	St 1 | St 2 | St 3 | St 4		
10						ESWT | HILT | TECAR | BFR | Cryo | ES | Cup	Maitland | Mulligan | DN | IASTM	St 1 | St 2 | St 3 | St 4		
11						ESWT | HILT | TECAR | BFR | Cryo | ES | Cup	Maitland | Mulligan | DN | IASTM	St 1 | St 2 | St 3 | St 4		
12						ESWT | HILT | TECAR | BFR | Cryo | ES | Cup	Maitland | Mulligan | DN | IASTM	St 1 | St 2 | St 3 | St 4		

DISCHARGE & MILESTONE SUMMARY

FINAL OUTCOME

PROM / OBJECTIVE METRIC	INITIAL BASELINE	DISCHARGE SCORE	Δ CHANGE	MCID MET?	CUTOFF CLEARED?
1. FAAM – ADL / Sports	____% / ____%	____% / ____%	+____%	☐	☐
2. VISA-A (Achilles)	____ / 100	____ / 100	+____	☐	☐
3. CAIT (Instability)	____ / 30	____ / 30	+____	☐	☐
4. LEFS (Lower Extremity)	____ / 80	____ / 80	+____	☐	☐

Physician Follow-Up Sign-Off Name _____ Signature _____ Date _____

OBJECTIVE RE-TEST MILESTONES

MEASURE	BASE	WK 4	WK 8	WK 12	D/C
WBLT (cm) L/R					
SLHR reps L/R					
Y-Balance anterior (cm) L/R					
Hop LSI %					

Discharging Clinician _____ **Sig.** _____

GO TO 1 2 3 4 5 6 7 8 9